

Extracting Data from PDFs

Week 9 · Documents module · Textbook Chapter 9

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This week at a glance

PDFs are the web's filing cabinets. This week we pry official records out of them and turn locked documents into data.

Monday | Concepts & notebook intro

- Why PDFs resist extraction; reconstruction vs. reading

Wednesday | Notebook lab

- Work the Chapter 9 notebook: PyPDF, regex cleanup, Boulder City Council minutes

Friday | Textbook revisions

- Propose & peer-review pull requests improving Chapter 9

Companion notebook

`ch-09-pdfs.ipynb`

Bring a laptop

Wednesday is hands-on — we extract and debug live.

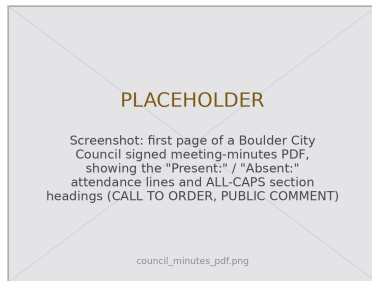
1. Monday • Concepts

The web's filing cabinets

PDFs are everywhere in institutional record-keeping: council minutes, bill texts, financial reports, court filings, academic papers, environmental impact assessments.

They are the format of choice for “official” documents because they **preserve visual layout** across any device or printer. But that priority — visual fidelity over semantic structure — makes them **hostile to programmatic extraction**.

The records inside are *readymade* data (Salganik (2018)). Somebody made them for administration, not for research. With effort you can still use them to answer questions that their authors never expected.



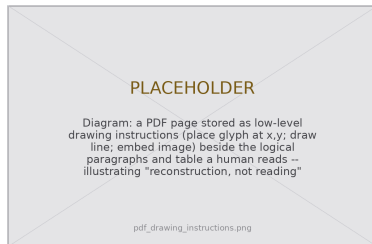
Boulder City Council minutes — our target this week.

A computer never reads a PDF

When **you** read a PDF you see paragraphs, tables, headers, and page numbers.

A **computer** reads a different thing. It sees a sequence of drawing instructions: *put this character at (x, y) . Draw a line from here to there. Add this image.*

So text extraction is **reconstruction, not reading** — the computer must infer logical structure from spatial layout. The results will not be perfect, but they can be good enough to turn inaccessible records into analyzable datasets.



Glyphs at coordinates, not sentences — structure is inferred.

Soft enclosure & the infrastructure of transparency

Some governments publish minutes, budgets, and filings *only* as PDFs. The post-API chapter calls this **soft enclosure**. The records are public in name. In practice you cannot analyze them.

Building extraction skills enacts the **oversight** and **ownership** values — public records belong to the public, and technical barriers are themselves a form of enclosure.

Adobe created the PDF in 1993 for visual fidelity across devices — at the cost of semantic accessibility. **DocumentCloud** (ProPublica & IRE), **MuckRock** (automating FOIA), and the **Sunlight Foundation** built the infrastructure that bridges public documents and accessible data.



DocumentCloud: shared infrastructure for analyzing document dumps.

This week's companion notebook builds one skill at a time:

- **PyPDF basics** — open a file, read metadata, extract page-level text (and see how messy it is)
- **Cleanup** — normalize whitespace *without* destroying line structure; use regex to impose fields
- **Case study** — Boulder City Council minutes: council attendance and public-comment counts
- **Pipeline** — a reusable class that processes a whole corpus of PDFs into a DataFrame

Connecting to Ch. 7 & the post-API theme

Once text is out, the **bag-of-words** preprocessing from the archives chapter (Ch. 7) sharpens topic tracking across many meetings. And PDF-only publishing *is* the “soft enclosure” the post-API chapter warns about — extraction is how we push back.

2. Wednesday · Notebook Lab

pypdf gives you document metadata and page-level text. Start by looking at what actually comes out:

```
from pypdf import PdfReader

reader = PdfReader("council_minutes_2024_01.pdf")
print(len(reader.pages), "pages")
print(reader.metadata) # title, author, creation date...

page = reader.pages[0]
text = page.extract_text()
print(text[:500]) # expect mess: headers/footers mixed in, tables garbled,
                  # line breaks that don't match paragraphs
```

The one line that flattens the page

Tempting one-liner:

```
# DON'T: \s matches newlines too,  
# so this flattens the whole page  
re.sub(r'\s+', ' ', raw_text)
```

Correct — collapse spaces & tabs, **keep** the line breaks:

```
text = re.sub(r'[\t ]+', ' ', raw_text)  
lines = [ln.strip() for ln in text.split('\n')]  
clean = '\n'.join(ln for ln in lines if ln)
```

The most important choice is what *not* to collapse.

Line structure is real information:
section headings are recognizable
**precisely because they sit alone
on their own lines.**

Erase the newlines and no heading
detector can ever find them.

Structured data hides inside a wall of characters; regex carves it out. Five patterns cover most government PDFs:

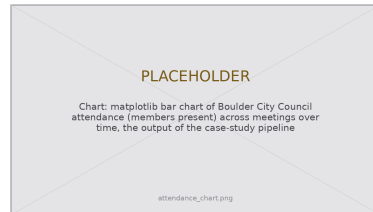
```
dates = re.findall(r'\d{1,2}/\d{1,2}/\d{2,4}', text) # 1/15/2024, 12/03/23
amounts = re.findall(r'\$[\d,]+\.\d*', text) # $1,250.00, $3,400,000
emails = re.findall(r'[\w.+-]+@[ \w-]+\.[\w.-]+', text) # clerk@boulder.colorado.gov

# ALL-CAPS section headings on their own line (a literal space, NOT \s,
# so one match can't swallow several consecutive heading lines):
headings = re.findall(r'^[A-Z][A-Z0-9 .:&-]{5,}$', text, re.MULTILINE)
# -> ['CALL TO ORDER', 'PUBLIC COMMENT', 'CONSENT AGENDA', 'ADJOURNMENT']
```

Each pattern encodes an assumption (slash dates, a leading \$, uppercase headings). **Test across multiple pages and documents** before trusting a pipeline. New to regex? See *Missing Manual* Ch. 18.

Case study — council attendance

```
def extract_attendance(pdf_path):
    text = PdfReader(pdf_path).pages[0].extract_text()
    # names follow "Present:" up to "Absent:" or end of line
    present = re.search(r'Present:\s*(.*?)\s*(?:Absent:|$)',
                        text, re.IGNORECASE | re.MULTILINE)
    absent = re.search(r'Absent:\s*(.*)$',
                       text, re.IGNORECASE | re.MULTILINE)
    return {"present": split_names(present),
            "absent": split_names(absent)}
```



Counts across meetings become a trend — scale reveals
what one PDF can't.

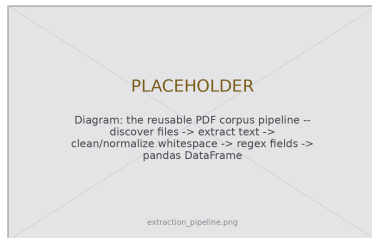
Be honest about limits: “Absent: None” becomes a phantom member named `None`; titles and separators vary. Treat output as a **first draft** — spot-check and patch.

Scale it — a reusable corpus pipeline

Wrap the workflow so one class processes a whole directory into a DataFrame:

```
class PDFCorpusAnalyzer:
    def __init__(self, directory, cleanup_patterns=None):
        self.files = sorted(f for f in os.listdir(directory)
                             if f.lower().endswith(".pdf"))
        self.cleanup_patterns = cleanup_patterns or []

    def process_all(self, extract_fn):
        rows = []
        for f in self.files:
            try: # one bad PDF won't kill the run
                text = self.extract_text(f)
                rows.append(**extract_fn(f, text), "file": f)
            except Exception as e:
                rows.append({"file": f, "error": str(e)})
        return pd.DataFrame(rows)
```



Discover → extract → clean → regex fields →
DataFrame.

PyPDF handles text-based PDFs well, but two common cases defeat it.

Tables — PyPDF flattens them into a text stream, losing rows and columns. `pdfplumber` preserves the grid:

```
import pdfplumber
with pdfplumber.open("report.pdf") as pdf:
    table = pdf.pages[0].extract_table() # a list of row-lists (cells)
    df = pd.DataFrame(table[1:], columns=table[0])
```

Scanned PDFs — a scan has no text layer, so PyPDF returns empty or garbled output. You need **OCR**:

```
# pytesseract + the Tesseract engine turn page images into text;
# accuracy varies with scan quality. (camelot is another table option.)
```

Know your tool's limits: reach for `pdfplumber`, `camelot`, or OCR when the layout demands it.

Lab: work these, then show-and-tell

Work in pairs. Do these exercises from the notebook:

1. Find one element that occurs in 10 or more government PDFs. Extract it. Put the results in a DataFrame.
2. Compare the quality of PyPDF on three documents: a text report, a document with many tables, and a slide deck.
3. Extract a budget table with `pdfplumber`. Extract it again with `pypdf`. Make numeric columns from the result.
4. Select a term such as “housing” or “police.” Count it in one year of meeting minutes. Plot the spikes. Use the bag-of-words method from Ch. 7.
5. Compare `pdfplumber` and `pypdf` on the same PDF.
6. **5617**: measure the extraction quality for the PDFs of two cities. Write a memo about data quality. Use the categories of Gebru et al. (2021)

Show-and-Tell

Bring one of these to class:

- a bug that you could not solve
- data that you collected and find interesting
- a question for a research design

PDFs are built for eyes, not for code.

Every extraction is an educated reconstruction —
never perfect, often good enough.

Imprecision is the price of scale —
and scale is what turns documents into data.

3. Friday • Textbook Revisions

Today we make the book better. Each of you opens a **pull request** against `ch-09-pdfs.qmd` and reviews a classmate's.

The workflow (from Week 1):

1. Branch / fork the book repo
2. Edit the chapter; commit with a clear message
3. Open a PR describing *what* and *why*
4. Review two peers' PRs: kind, specific, actionable



High-value targets — pick one and scope it to a single PR:

- **Run it against real Boulder minutes.** The # Returns: outputs and five regex patterns are *illustrative*. Download actual signed minutes, report what the patterns catch and miss, and label sample output as representative.
- **Add the chapter's first figures.** It is all code and prose today: a “drawing-instructions vs. logical text” diagram, a pipeline flowchart, or an annotated minutes screenshot would anchor Monday's core idea.
- **Harden `extract_attendance`.** The chapter itself flags “Absent: None,” title variants (“Council Member Smith”), and semicolon / line-break separators. Contribute handling **plus tests** for the formats you actually hit.
- **Make an exercise self-checking.** Add expected output, a rubric, or a starter cell to the open-ended “find and extract” and “topic tracking” exercises.
- **Deepen “When PyPDF Is Not Enough.”** Turn the two-line `pdfplumber` / OCR sketches into a runnable table-extraction or Tesseract example tied to “Further Reading.”

A good revision, and a good review

A strong PR

- One focused change, clearly titled
- Explains the problem it fixes
- Builds (`quarto render`) without errors
- Matches the book's voice and notation
- Discloses any AI assistance

A strong review

- Runs / reads the change, not just skims
- Names specifics (line, claim, code)
- Separates “must fix” from “nice to have”
- Is generous — assume good faith
- Ends with a clear verdict

Next week: a new module on **APIs** — when data arrives as JSON and no parsing is required (starting with Wikipedia).

Key takeaways

1. PDFs optimize **visual fidelity over structure** — extraction is reconstruction, not reading.
2. `pypdf` gives page-level text & metadata; expect messy output and plan to clean it.
3. Collapse spaces/tabs but **keep newlines** — line structure is information; regex imposes the rest.
4. Abstract into a **pipeline** so one bad file doesn't kill a corpus run; test patterns across documents.
5. Know PyPDF's limits: `pdfplumber` for tables, OCR for scans. Imprecision is the price of scale.